

Problem 3 (09AMO2)

The answer is n for all even n and $n + 1$ otherwise.

The construction for even n is $\{1, \dots, n\}$.

The construction for odd $n = 2k + 1$ is $\{-2k - 1, \dots, -k - 1\} \cup \{k + 1, \dots, 2k + 1\}$.

FTSOC let there be a subset S satisfying the conditions where $|S|$ is greater than the given bounds.

Let m be the minimal absolute value over all terms in S . By Pigeonhole, $m \leq \left\lfloor \frac{n}{2} \right\rfloor$. Note that $m = 0$ is also impossible since then we can take $a = b = c = 0$.

One of $m, -m$ is in S to achieve the minimum. WLOG, let $m \in S$. For each of the pairs,

$$(-n, n - m), (-n + 1, n - m - 1), \dots, (-2m, m)$$

It is impossible for both elements of the pair to simultaneously be in S , since taking those two terms and m would sum to zero. Therefore, we are guaranteed $n - 2m + 1$ terms from the range $[-n, -2m] \cup [m, n]$ do not exist.

None of the $2m - 1$ terms from $[-m + 1, m - 1]$ may exist either.

In total, n terms do not exist, leaving at most $n + 1$ terms in S .

The only contradiction case is if $|S| = n + 1$ and n is even. The equality case must hold meaning among every pair from,

$$(-n, n - m), (-n + 1, n - m - 1), \dots, (-2m, m)$$

Exactly one of the two be in S . Every term on $(-2m, -m]$ also $\in S$ meaning $-m \in S$. Hence, in each of the following pairs,

$$(m - n, n), \dots, (-m, 2m)$$

We symmetrically get exactly one of them is in S . Since every term on $(-2m, -m] \in S$, everything from $[2m, \min(4m - 2, n)] \notin S$.

We may induct to get what $|x| \in S$ or not... Prove that among $|x| \leq 2k$ there's always at least $2k + 1$ excluded.

Problem 4 (14PTNMA4)

Let sequence $p_0, p_1, \dots$ be defined such that p_i is the probability that $X = i$.

We claim the minimum is $p_0 = \frac{1}{3}$, achieved with $p_1 = \frac{1}{2}, p_2 = 0, p_3 = \frac{1}{6}$.

Note,

$$\sum_{n \geq 1} np_n = 1 \quad (a)$$

$$\sum_{n \geq 1} n^2 p_n = 2 \quad (b)$$

$$\sum_{n \geq 1} n^3 p_n = 5 \quad (c)$$

Therefore,

$$\sum_{n \geq 1} \left(\frac{1}{6}n^3 - n^2 + \frac{11}{6}n \right) p_n = \frac{2}{3}$$

Note that $\frac{1}{6}n^3 - n^2 + \frac{11}{6}n = \frac{(n-1)(n-2)(n-3)}{6} + 1$. Therefore,

$$\frac{2}{3} = \sum_{n \geq 1} \left(\frac{1}{6}n^3 - n^2 + \frac{11}{6}n \right) p_n \geq p_1 + p_2 + p_3 + \dots$$

Since $p_0 + p_1 + \dots = 1$, we are done with $p_0 \geq \frac{1}{3}$.

8 clubs; 2.5 hours

Problem 5 (09USATST7)

We claim the only solution is $(x, y, z) = (2, 4, 6)$

Plug in $x = a + 2; y = b + 4; z = c + 6$. We may plug this in to get,

$$a^3 + 6a^2 + 9a = a(a + 3)^2 = -12b$$

$$b^3 + 6b^2 + 9b = b(b + 3)^2 = 3c$$

$$c^3 + 6c^2 + 9c = c(c + 3)^2 = 27a$$

If any of a, b, c is zero, observe that $a = b = c = 0$ immediately follows. Assume FTSOC there is a set of solutions with non-zero a, b, c . Multiply the three equations to get,

$$abc(a + 3)^2(b + 3)^2(c + 3)^2 = (-12 \cdot 81)abc$$

$$\Leftrightarrow (a + 3)^2(b + 3)^2(c + 3)^2 = (-12 \cdot 81)$$

This is impossible by trivial inequality.

Problem 6 (00IMO4)

The answer is 12. There are 6 cases where the red, green, and blue boxes claim exactly one residue class mod 3. There are 6 cases where 1, 100 go into two different boxes and 2, ..., 99 in a third box.

We look at the smallest possible d such that $x, x + d$ are in the same bucket.

Case 1: $d = 1$

WLOG, let $n, n + 1$ both go into the red bin. Note that we cannot have consecutive numbers $k, k + 1$ go into the green and blue bins in some order in this case because if the given sum is $n + k + 1$, we cannot know if it is $n + (k + 1)$ or $(n + 1) + k$.

If FTSOC there are two consecutive numbers $k, k + 1$ going into another colored bin, WLOG green, then consecutive numbers cannot go into red and blue. Therefore, no numbers go into blue or everything goes into blue.

Therefore, if FTSOC $2 \leq n \leq 99$ is in the green or blue bin (WLOG green), then $n \pm 1$ are in the red bin. For whichever number k appears in the blue bin, then at least one of $k \pm 1$ is in 1, ..., 100 then we have two ambiguous ways to make $n + k \pm 1$.

In this case, 1, 100 must go into two different boxes and 2, ..., 99 must all go in a third box.

Case 2: $d = 2$

WLOG let $n, n + 2$ go into Red and $n + 1$ go into Green. If $n + 3$ or $n - 1$ go into Blue, we have an issue with ambiguous sums $2n + 3$ or $2n + 1$. We may hence induct both ways to get blue never appears. No cases.

Case 3: $d \geq 3$

By Pigeonhole the only way this is possible is if red, blue, and green bins each claim a different residue class modulo 3.

14 clubs; 4.5 hours

Problem 7 (19CHN1)

We claim the minimum is -512 achieved with $-1, -1, -1, -1, 9$ and maximum is 288 achieved with $-1, -1, -1, 4, 4$.

Let $A = a + b; B = c + d; C = e + a; D = b + c; E = d + e$.

$$a + b + c + d + e = 5 \Rightarrow A + B + C + D + E = 10$$

$$a, b, c, d, e \geq -1 \Rightarrow A, B, C, D, E \geq -2$$

Furthermore, we may check that,

$$A, B, C, D, E \leq 8; A + B, B + C, \dots, E + A \leq 6$$

FTSOC, if not, and WLOG $A = a + b > 8$, then $c + d + e = 5 - (a + b) < -3$. Similar argument to show $A + B, \dots \leq 6$.

We show that these conditions are sufficient to show $-512 \leq S = ABCDE \leq 288$.

To prove the minimum, we casework on the number of negatives among A, B, C, D, E .

Case 1: 1 negative. The positives sum to at most 12 because the negative is at least -2 and the total sum is 10. The minimum product is $3 \cdot 3 \cdot 3 \cdot 3 \cdot (-2) = -162$ by AM-GM.

Case 2: 3 negatives. The two positives sum to at most 16 because the three negatives are at least -6 . The product of the negatives is at least -8 and the product of the positives at most 64 by AM-GM. Therefore, the total product at least -512 .

Case 3: 0, 2, 4, 5 negatives. Either the product is nonnegative, or it is impossible for the sum to be positive.

To prove the maximum, we similarly casework on number of negatives.

Case 1: 0 negatives. By AM-GM, the maximum product is $2^5 = 32$.

Case 2: 2 negatives. By Pigeonhole, of the 3 positives, at least two of them are adjacent. WLOG, let $a, b \geq 0$. Note that $a + b \leq 6$. Since $-2 \leq c, d, e \leq 8$, by AM-GM, the max product $abcde$ for three positives is $3 \cdot 3 \cdot (-2) \cdot 8 \cdot (-2) = 288$.

Case 3: 4 negatives. The positive number is at most 18 and the negatives at least -2 . One bound we can get is $(-2)^4 \cdot 18 = 288$ (though this is not attainable).

Case 4: 1, 3, 5 negatives. The total product cannot be positive in this case.

Problem 9 (16EGMO5)

Since $n \leq 2k - 1$, it is impossible for there to be more than one of each horizontal block in each row or vertical block in each column. To minimize the number of tiles, we maximize the number of rows and columns with no aligned tiles.

Let there be r rows and c columns containing no tiles in the direction. WLOG, let $r \leq c$.

Main claim: If $c \geq k$, then $r = 0$.

Assume $c \geq k$. We mark the c columns without a vertical block red.

Lemma: If column m is not red but column $m + 1$ is, then column $m + k \leq n$ are all red; if column m is not red but column $m - 1$ is, then column $m - k \geq 1$.

Proof: We prove the first condition and the second follows symmetrically. Assume that there is a vertical block in column m . Consider the image of block you get by translating this block right by

one unit – one of the k cells must be filled. Since column $m + 1$ is red, it must be filled with a horizontal block. For the horizontal block to fit, $m + k \leq n$ must be satisfied. $\square$

FTSOC, if the red columns do not form a single continuous rectangle, then there must be two indices a, b where $a - 1$ is red and a is not red; b is not red and $b + 1$ red. The lemma cannot be satisfied by size. (!)

Given that the red columns form a single continuous rectangle of width $\geq k$, and given that there are no more vertical blocks filling this rectangle, we must place n horizontal blocks in each row of the rectangle meaning $r = 0$ as desired.

24 clubs; 7 hours

Problem 10 (19EGMO2)

Sketch:

Claim answer $\frac{n(n+1)}{2}$

Consider the eight cells in the SOI of each block that no other blocks may occupy, we want to ensure the SOIs cover the board without overlapping and have as much spill outside as possible.

Construction looks like four pyramids stacked in each direction.

Proof by bounding how many spills over for each type of block on edges (also show vertically attaching to edges is inefficient)

Problem 13 (13SLA4)

Lemma 1: $a_1 \leq n$.

Proof: If FTSOC claim is false, then by non-strictly increasing sequence condition $a_i > n$ for all i meaning it is impossible for $a_{a_1} \leq n + 1 - 1 = n$. $\square$

As a result, $a_i \leq 2n$ for all i .

Lemma 2: $a_{a_i-n} \leq n$

Proof: We first show that $a_{a_i-n} > n$ is not possible.

FTSOC assume $a_{a_i-n} > n$.

If $a_1 \geq a_i - n$, then we have $a_{a_1} \geq a_{a_i-n}$ since $a_i, a_i - n \leq n$ and the sequence is non-strictly increasing. This is impossible because $a_{a_1} \geq a_{a_i-n} > n$ is not allowed as $a_{a_1} \leq n$. (!)

If $a_1 < a_i - n$, then $a_i \leq a_n \leq a_1 + n$ cannot be true either. (!) $\square$

Now having shown

Claim 1: If $a_i > n$, then $a_{a_i-n} < i$.

Proof: FTSOC let this not be true and $a_{a_i-n} \geq i$. Since we know $a_{a_i-n} \leq n$ and $i \leq n$, the non-strictly increasing condition gives,

$$a_{a_{a_i-n}} \geq a_i$$

However,

$$a_{a_{a_i-n}} \leq n + (a_i - n) - 1 \quad (!)$$

We have a contradiction and the claim is proven. $\square$

Claim 2: There are at most $\max(0, n - a_i)$ terms greater than or equal to $n + i$ for $i = 1, \dots, k$

Proof: The case where $a_i > n$ follows from claim 1. If $a_i \leq n$ and this was not the case, then $a_{a_i} \geq n + i$ which is a contradiction. $\square$

Plug in $b_i := a_i - n$. The claim states that there are at most $\max(0, -b_i)$ terms greater than or equal to i out of all b_i . We want to show $b_1 + \dots + b_n \leq 0$.

Let k be the largest integer such that $b_k \leq 0$. It suffices to show that,

$$\begin{aligned} -b_1 - \dots - b_k &\geq b_{k+1} + \dots + b_n \\ \Leftrightarrow -b_1 - \dots - b_k &\geq \sum_{i=1}^{\infty} \text{Number of } b_{k+1}, \dots, b_n \text{ which are at least } i \end{aligned}$$

This follows from claim 2.

34 clubs; 9 hours

Problem 14 (14TWNQ3J5)

To prove the lower bound,

$$\frac{1}{n} \left(\sum_{i=1}^n \frac{x_{i-1} + x_{i+1}}{x_i} \right) = \frac{\left(\frac{x_1}{x_2} + \frac{x_2}{x_3} + \dots + \frac{x_n}{x_1} \right) + \left(\frac{x_2}{x_1} + \frac{x_3}{x_2} + \dots + \frac{x_1}{x_n} \right)}{n}$$

By AM-GM, this is at least 2 as desired.

We induct on n to show $k_1 + \dots + k_n \leq 3n - 1$ for all $n \geq 1$.

Base case: $n = 1$. Trivially, $k_1 = \frac{x_0 + x_2}{x_1} = \frac{2x_1}{x_1} = 2 = 3(1) - 1$.

Inductive step: WLOG let x_n be the maximum x_i . We claim that if we pop x_n from the sequence, the newly created sequence still satisfies all k_i are integers, and the sum $\sum k_i$ decreases by at most 3. This suffices by the inductive hypothesis.

By removing x_n , none of $k_2, \dots, k_{n-2}$ are changed. The value k_n is gone.

If $\frac{x_{n-1}+x_1}{x_n} = k_n = 2$, then we must have $x_{n-1} = x_n = x_1$. We see that k_{n-1} and k_1 are also unchanged in this case. In total, k_i decreases by 2 and everything is still an integer.

If $\frac{x_{n-1}+x_1}{x_n} = k_n = 1$, then $x_{n-1} + x_1 = x_n$. Each of k_{n-1} and k_1 decreases by 1.

k_n cannot be any other value since x_n is maximum so we are done. $\square$

Problem 15 (16SLC5)

We claim that the answer is $n - 3$ when n is odd and $n - 2$ when n is even. Label the vertices of the n -gon $A_1, \dots, A_n$. The construction (sketch) is drawing two intersecting diagonals and filling in the rest with non-intersecting diagonals.

When n is odd, no two diagonals may intersect inside the n -gon since no two diagonals are perpendicular (this can be proven using complex numbers). As a result, we may induct on n to get the answer is $n - 3$. (Sketch) the construction is trivial.

Color all the diagonals which intersect another diagonal inside the n -gon red.

Lemma: All the red diagonals are either parallel or perpendicular to each other.

Proof: FTSOC let this not be the case and we have two disjoint pairs of intersecting perpendicular diagonals. After the first pair of diagonals are drawn, the vertices are separated into four groups of at most $\frac{n}{2}$ vertices and there is no space for another intersection. $\square$

Imagine we draw in all the red diagonals first, then draw in the remaining diagonals. After the red diagonals are done, no more intersections inside the n -gon may exist. We now calculate how many diagonals

Main Claim: If we have k red diagonals and m vertices on the n -gon where two red diagonals meet, the maximum number of diagonals we can draw in total is at most $n - k + m$.

We prove this by induction on the diagonals... (since this is a sketch detail omitted; base case is two diagonals that intersect inside)

To prove the bound, it now suffices to show that $k \geq m + 2$. Since the red diagonals are perpendicular to each other, at most two can meet at a single vertex. On the other hand, each diagonal goes through two vertices. We know $k - m$ is even and $k \geq m$ as a result, meaning it suffices to prove $k = m$ is impossible.

FTSOC, let there be a configuration of red diagonals where $k = m$. Take one of the diagonals in this configuration with minimal length. Call this diagonal d_1 . We now show that there is a different diagonal shorter than d_1 to complete the contradiction.

Since d_1 is red, there must be a diagonal d_2 perpendicular to d_1 intersecting it inside the n -gon. d_1 splits the n -gon into two halves, and we consider the end of d_2 in this smaller half. Since $k = m$, there must be another diagonal d_3 passing through this vertex. $d_3 \perp d_2 \perp d_1$ so $d_3 \parallel d_1$ and hence the length of d_3 is smaller than that of d_1 as desired.

48 clubs; 12 hours

Problem 16 (23USAEST2)

We claim the maximum is 7993, achieved with,

$$f(0) = 48; f(1) = f(2) = \dots = f(42) = 47$$

$$g(0) = 96; g(1) = 95; g(2) = g(3) = \dots = g(84) = 94$$

Note $g(x) \leq 2f\left(\frac{x}{2}\right)$ when x is even and $g(x) \leq f\left(\frac{x-1}{2}\right) + f\left(\frac{x+1}{2}\right)$ when x is odd.

Note $g(x) \leq f(0) + f(x)$ when $x \leq 42$ and $g(x) \leq f(x-42) + f(42)$ when x is odd.

Combining these equations,

$$g(x) \leq \left(2 - \frac{2}{43}\right) f\left(\frac{x}{2}\right) + \frac{1}{43}(f(0) + f(x)) \text{ when } x \leq 42 \text{ is even}$$

$$g(x) \leq \left(1 - \frac{1}{43}\right) \left(f\left(\frac{x-1}{2}\right) + f\left(\frac{x+1}{2}\right)\right) + \frac{1}{43}(f(0) + f(x)) \text{ when } x \leq 42 \text{ is odd}$$

$$g(x) \leq \left(2 - \frac{2}{43}\right) f\left(\frac{x}{2}\right) + \frac{1}{43}(f(x-42) + f(42)) \text{ when } x > 42 \text{ is even}$$

$$g(x) \leq \left(1 - \frac{1}{43}\right) \left(f\left(\frac{x-1}{2}\right) + f\left(\frac{x+1}{2}\right)\right) + \frac{1}{43}(f(x-42) + f(42)) \text{ when } x > 42 \text{ is odd}$$

Adding this up,

$$g(0) + g(1) + \dots + g(84) \leq \left(3 + \frac{41}{43}\right)(f(0) + \dots + f(42)) \leq \frac{2022 \cdot 170}{43}$$

$$= \frac{2021}{43} \cdot 170 + \frac{170}{43} = 47 \cdot 170 + \frac{170}{43} = 7993 + \frac{41}{43} < 7994$$

Since $g(0) + \dots + g(84)$ is a nonnegative integer, we are done.

Problem 19 (18SLC5)

Let player i arrive on day a_i and leave at day e_i . The amount Evan must pay is,

$$\sum_{i=1}^{2k} e_i - a_i + 1 = 2k + \sum_{i=1}^{2k} e_i - \sum_{i=1}^{2k} a_i$$

We claim the minimum is $\frac{k(4k^2+k-1)}{2}$, achieved by ordering the matches as follows (if x is the value of the cell in row r and column c indicates that players r, c play on day x).

X	1	2	4	7	11	14	16
1	X	3	5	8	12	15	17
2	3	X	6	9	13	18	19
4	5	6	X	10	20	21	22
7	8	9	10	X	23	24	25
11	12	13	20	23	X	26	27
14	15	18	21	24	26	X	28
16	17	19	22	25	27	28	X

In this labeling, we may check that the day that player n plays player m is strictly increasing as m increases from 1 to $2k$. *Sketch: Do some arithmetic to check answer is satisfied.*

There are $n - 1$ players before player n arrives. Thus, at most $\binom{n-1}{2}$ games occurred before then meaning,

$$a_n \leq 1 + \binom{n-1}{2}$$

Similarly,

$$e_n \geq \binom{2k}{2} - \binom{2k-n}{2}$$

Thus,

$$\sum_{n=1}^k e_{2k-n+1} - a_n \geq \sum_{n=1}^k \left(\binom{2k}{2} - \binom{n-1}{2} \right) - \left(1 + \binom{n-1}{2} \right)$$

$$= \sum_{n=1}^k \binom{2k}{2} - 2 \binom{n-1}{2} - 1$$

We now find a tighter upperbound limit of $(a_n - 1) + \left(\binom{2k}{2} - e_{2k-n+1} \right)$, the number of games played before the n -th arrival and n -th to last departure. There are once again at most $n - 1$ players before and after meaning $\binom{n-1}{2}$ games played before and after, but there are at least $2n - 2k - 2$ players overlapping, so by PIE, the limit is,

$$(a_n - 1) + \left(\binom{2k}{2} - e_{2k-n+1} \right) \leq 2 \binom{n-1}{2} - \left(2 \binom{n-1}{2} - 2k \right)$$

$$\Leftrightarrow e_{2k-n+1} - a_n \geq \binom{2k}{2} - 2 \binom{n-1}{2} - 1 + \left(2 \binom{n-1}{2} - 2k \right)$$

We may now bash out the algebra to finish (e.g. using generating functions).

62 clubs; 16 hours